Experiment:1

AIM: Build a responsive web application for shopping cart with registration, login, catalog and cart pages using CSS3 features, flex and grid.

SOLUTION:

Responsive design means building a website, it automatically adjusts to different screen sizes and devices - like mobile phones, tablets, laptops, and desktops.

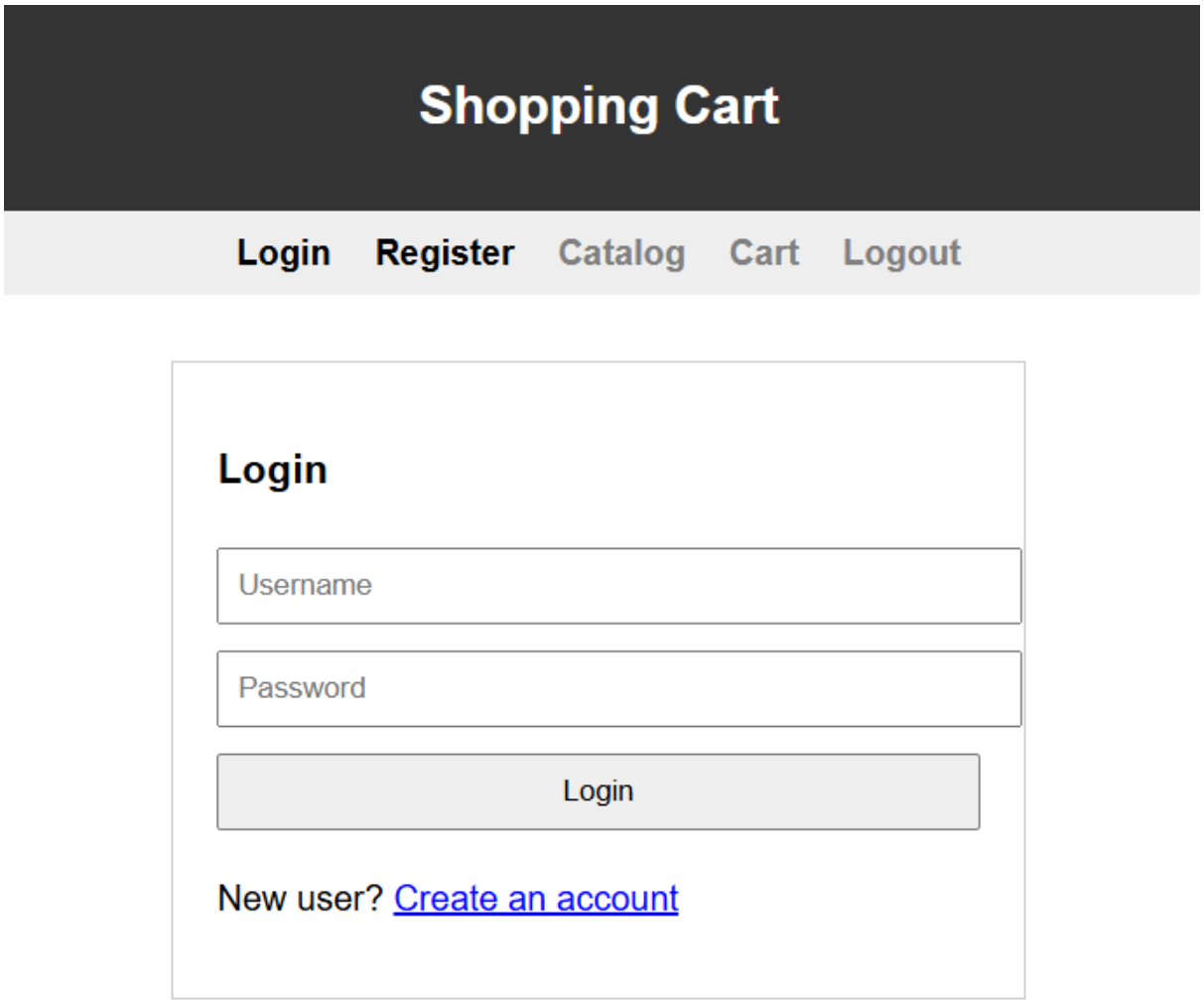
Source Code:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Login</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body onload="updateNav()">
<header><h2>Shopping Cart</h2></header>
<nav>
<a id="navLogin" href="index.html">Login</a>
<a id="navRegister" href="register.html">Register</a>
<a id="navCatalog" class="disabled" href="catalog.html">Catalog</a>
<a id="navCart" class="disabled" href="cart.html">Cart</a>
<a id="navLogout" class="disabled" href="logout.html">Logout</a>
</nav>
<div class="form-box">
<h3>Login</h3>
<input id="logUser" placeholder="Username">
<input id="logPass" type="password" placeholder="Password">
<button onclick="login()">Login</button>
<p id="logMsg"></p>
<p>New user? <span class="link"
onclick="location.href='register.html'">Create an account</span></p>
</div>
<script src="js/app.js"></script>
```

```
</body>
</html>
```

Output:

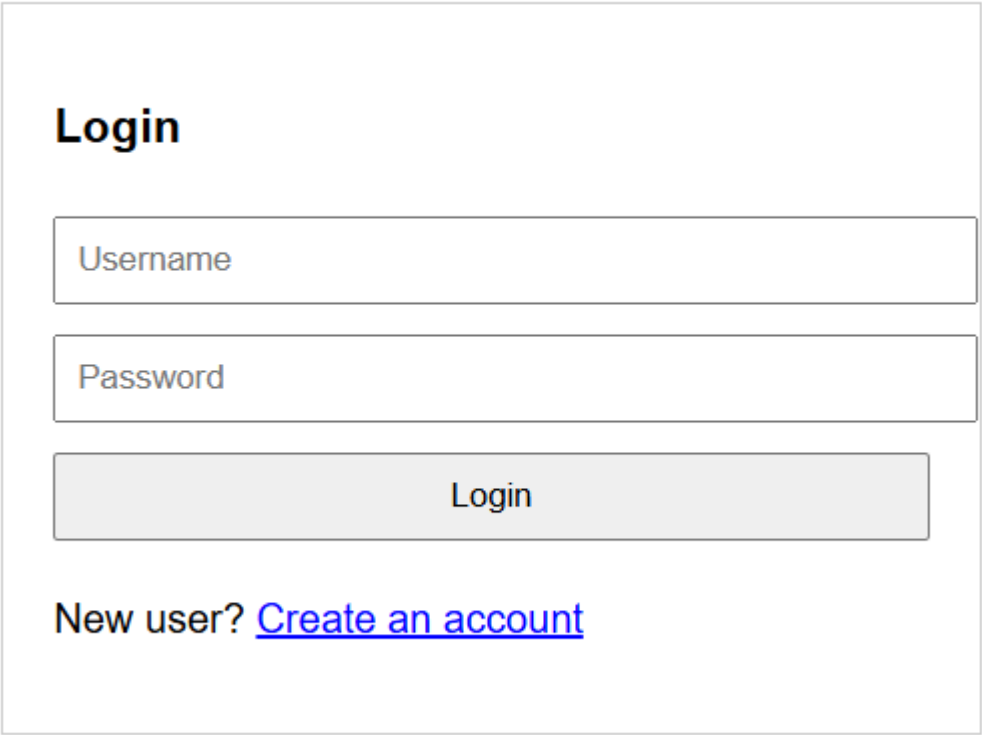


Login.html

```
<!DOCTYPE html>
<html>
<head>
<title>Login</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body onload="navSetup()">
<header><h2>Shopping Cart</h2></header>
<nav>
<a id="navLogin" href="login.html">Login</a>
<a id="navRegister" href="register.html">Register</a>
```

```
<a id="navCatalog" class="disabled" href="catalog.html">Catalog</a>
<a id="navCart" class="disabled" href="cart.html">Cart</a>
<a id="navLogout" class="disabled" href="logout.html">Logout</a>
</nav>
<div class="form-box">
<h3>Login</h3>
<input id="logUser" placeholder="Username">
<input id="logPass" type="password" placeholder="Password">
<button onclick="login()">Login</button>
<p id="logMsg"></p>
<p>New user? <span class="link" onclick="location.href='register.html'">Create an account</span></p>
</div>
<script src="js/app.js"></script>
</body>
</html>
```

Output:

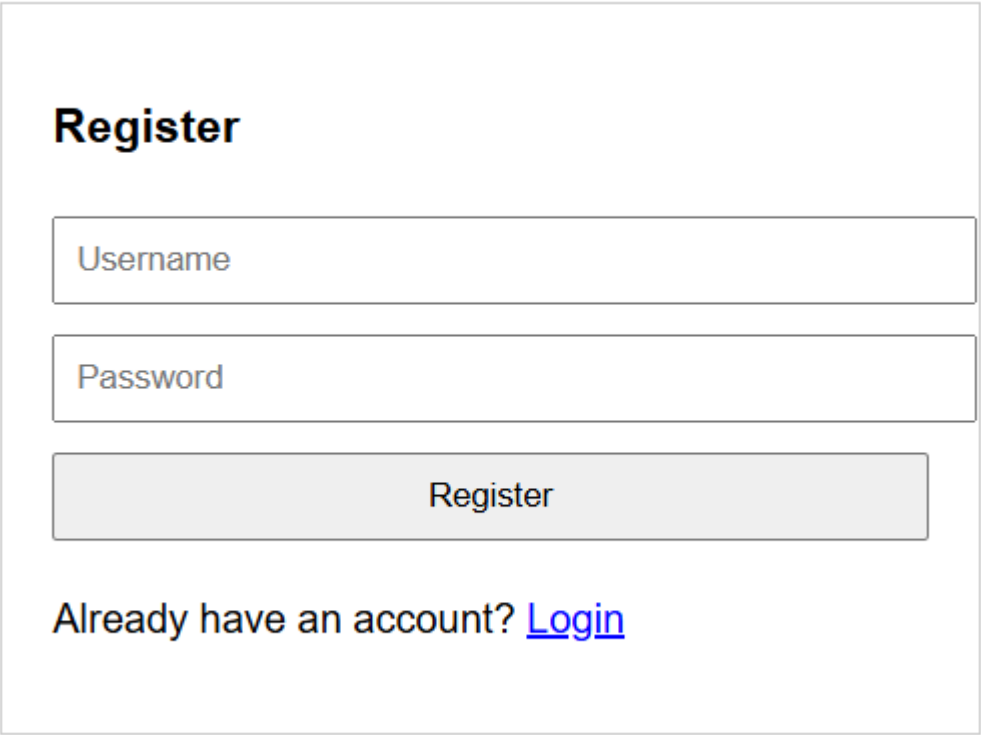


Register.html

```
<!DOCTYPE html>
```

```html
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Register</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body onload="updateNav()">
<header><h2>Shopping Cart</h2></header>
<nav>
<a id="navLogin" href="index.html">Login</a>
<a id="navRegister" href="register.html">Register</a>
<a id="navCatalog" class="disabled" href="catalog.html">Catalog</a>
<a id="navCart" class="disabled" href="cart.html">Cart</a>
<a id="navLogout" class="disabled" href="logout.html">Logout</a>
</nav>
<div class="form-box">
<h3>Register</h3>
<input id="regUser" placeholder="Username">
<input id="regPass" type="password" placeholder="Password">
<button onclick="register()">Register</button>
<p id="regMsg"></p>
<p>Already have an account?
<span class="link" onclick="location.href='index.html'">Login</span>
</p>
</div>
<script src="js/app.js"></script>
</body>
</html>
```

Output:

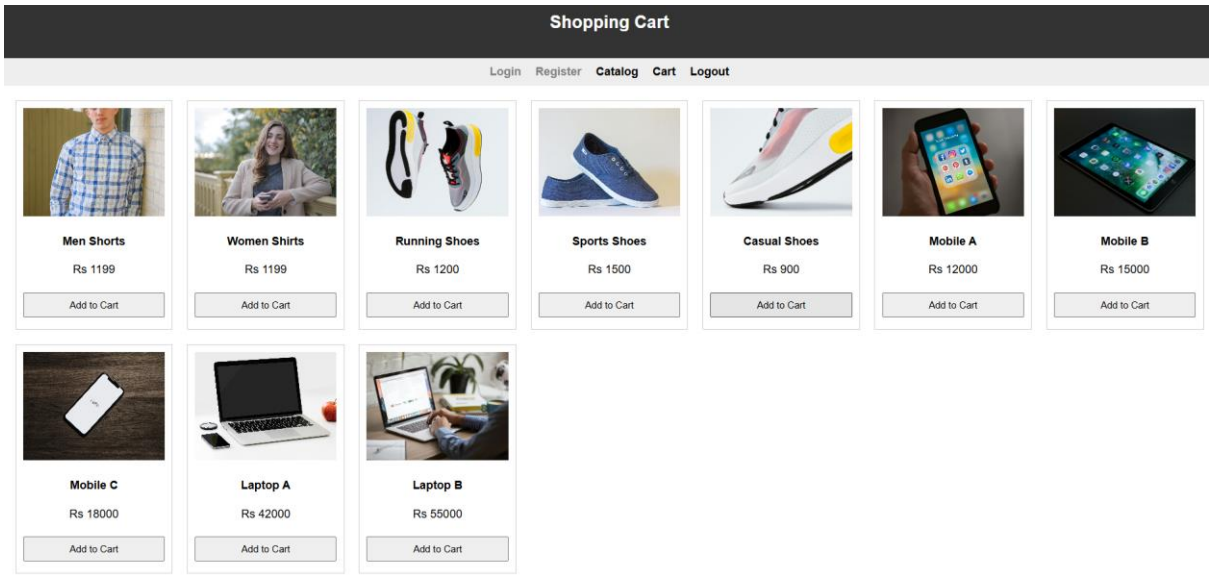


Catalog.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Catalog</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body onload="updateNav(); loadProducts();">
<header><h2>Shopping Cart</h2></header>
<nav>
<a id="navLogin" href="index.html">Login</a>
<a id="navRegister" href="register.html">Register</a>
<a id="navCatalog" class="disabled" href="catalog.html">Catalog</a>
<a id="navCart" href="cart.html">Cart</a>
<a id="navLogout" href="logout.html">Logout</a>
</nav>
<div class="grid" id="productGrid"></div>
```

```
<script src="js/app.js"></script>
</body>
</html>
```

Output:

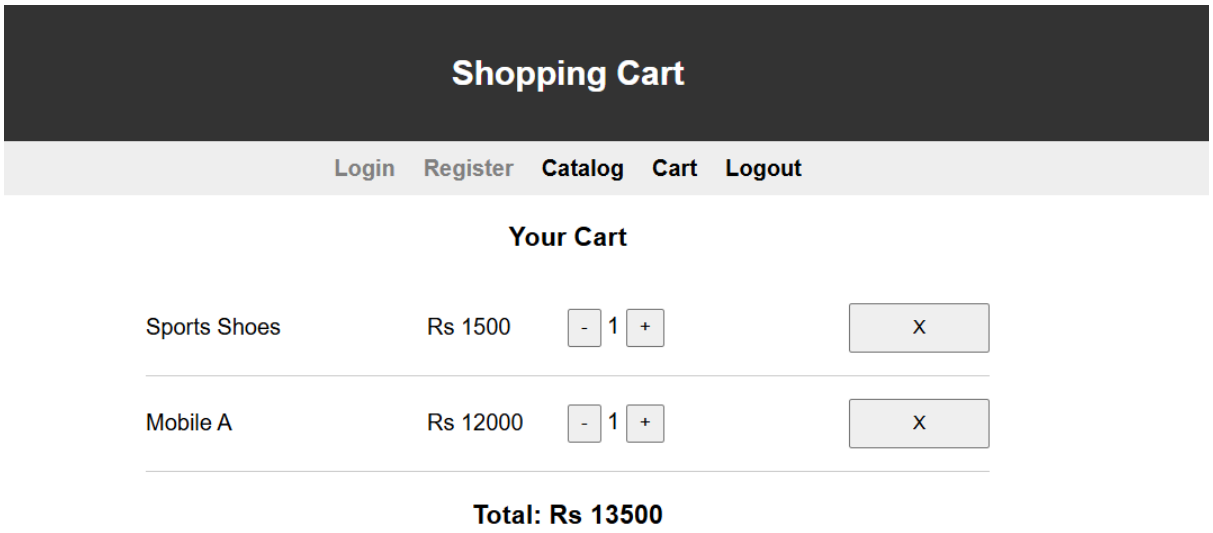


Cart.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Cart</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body onload="updateNav(); loadCart();">
<header><h2>Shopping Cart</h2></header>
<nav>
<a id="navLogin" href="index.html">Login</a>
<a id="navRegister" href="register.html">Register</a>
<a id="navCatalog" href="catalog.html">Catalog</a>
<a id="navCart" class="disabled" href="cart.html">Cart</a>
<a id="navLogout" href="logout.html">Logout</a>
</nav>
<h3 style="text-align:center">Your Cart</h3>
<div class="cart-box" id="cartList"></div>
```

```
<h3 style="text-align:center">Total: Rs <span id="totalAmount">0</span></h3>
<script src="js/app.js"></script>
</body>
</html>
```

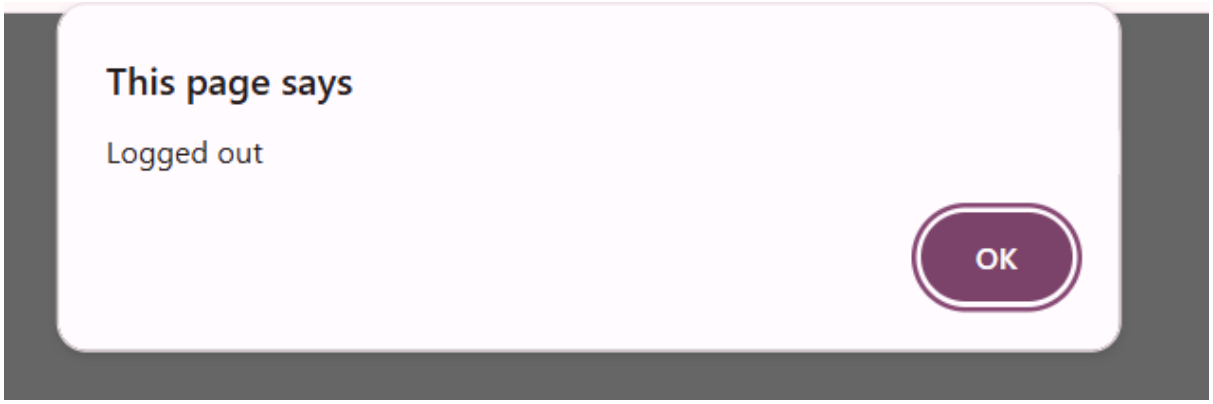
Output:



Logout.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Logout</title>
<link rel="stylesheet" href="css/style.css">
</head>
<body onload="logout()"></body>
<script src="js/app.js"></script>
</html>
```

Output:



Js/app.js

```
let isLoggedIn = localStorage.getItem("isLoggedIn") === "true";
let cart = JSON.parse(localStorage.getItem("cart")) || [];

// ---------- NAV CONTROL ----------
function updateNav() {
  if (isLoggedIn) {
    navRegister.classList.add("disabled");
    navLogin.classList.add("disabled");
    navCatalog.classList.remove("disabled");
    navCart.classList.remove("disabled");
    navLogout.classList.remove("disabled");
  } else {
    navRegister.classList.remove("disabled");
    navLogin.classList.remove("disabled");
    navCatalog.classList.add("disabled");
    navCart.classList.add("disabled");
    navLogout.classList.add("disabled");
  }
}

// ---------- REGISTER ----------
function register() {
  let username = regUser.value.trim();
  let password = regPass.value.trim();
  localStorage.setItem("user", username);
  localStorage.setItem("pass", password);
```

```
  regMsg.innerText = "Registered successfully. Please login.";
  setTimeout(()=>location.href="index.html",800);
}

// ---------- LOGIN ----------
function login() {
  if (
    logUser.value === localStorage.getItem("user") &&
    logPass.value === localStorage.getItem("pass")
  ) {
    isLoggedIn = true;
    localStorage.setItem("isLoggedIn","true");
    location.href="catalog.html";
  } else {
    logMsg.innerText = "Invalid credentials";
  }
}

// ---------- LOGOUT ----------
function logout() {
  localStorage.removeItem("isLoggedIn");
  localStorage.removeItem("cart");
  alert("Logged out");
  location.href="index.html";
}

// ---------- PRODUCTS ----------
let products = [
 {id:1,              name:'Men              Shorts',              price:1199,
img:'https://images.pexels.com/photos/769749/pexels-photo-769749.jpeg'},
 {id:2,             name:'Women             Shirts',             price:1199,
img:'https://images.pexels.com/photos/3775087/pexels-photo-3775087.jpeg'},
 {id:3,             name:'Running             Shoes',             price:1200,
img:'https://images.pexels.com/photos/2529148/pexels-photo-2529148.jpeg'},
```

```
 {id:4,             name:'Sports             Shoes',             price:1500,
img:'https://images.pexels.com/photos/19090/pexels-photo.jpg'},
 {id:5,             name:'Casual             Shoes',             price:900,
img:'https://images.pexels.com/photos/2529147/pexels-photo-2529147.jpeg'},
 {id:6,              name:'Mobile              A',              price:12000,
img:'https://images.pexels.com/photos/607812/pexels-photo-607812.jpeg'},
 {id:7,              name:'Mobile              B',              price:15000,
img:'https://images.pexels.com/photos/1334597/pexels-photo-1334597.jpeg'},
 {id:8,              name:'Mobile              C',              price:18000,
img:'https://images.pexels.com/photos/699122/pexels-photo-699122.jpeg'},
 {id:9,              name:'Laptop              A',              price:42000,
img:'https://images.pexels.com/photos/18105/pexels-photo.jpg'},
 {id:10,              name:'Laptop              B',              price:55000,
img:'https://images.pexels.com/photos/374074/pexels-photo-374074.jpeg'}
];

function loadProducts(){
  products.forEach(p => {
    productGrid.innerHTML += `
      <div class="item">
        <img src="${p.img}">
        <h4>${p.name}</h4>
        <p>Rs ${p.price}</p>
        <button onclick="addToCart(${p.id})">Add to Cart</button>
      </div>`;
  });
}

// ---------- CART ----------
function addToCart(id) {
  let item = cart.find(c => c.id === id);
  if (item) item.qty++;
  else {
    let p = products.find(p => p.id === id);
    cart.push({id:p.id,name:p.name,price:p.price,qty:1});
    alert(p.name+" added to cart");
```

```
  }
  localStorage.setItem("cart",JSON.stringify(cart));
}

function loadCart() {
  cartList.innerHTML="";
  let total=0;
  cart.forEach((c,i)=>{
    cartList.innerHTML+=`
      <div class="cart-item">
        <span>${c.name}</span>
        <span>Rs ${c.price}</span>
        <div>
          <button class="qty-btn" onclick="changeQty(${i},-1)">-</button>
          ${c.qty}
          <button class="qty-btn" onclick="changeQty(${i},1)">+</button>
        </div>
        <button onclick="removeItem(${i})">X</button>
      </div>`;
    total+=c.price*c.qty;
  });
  totalAmount.innerText=total;
}

function changeQty(i,val){
  cart[i].qty+=val;
  if(cart[i].qty<=0) cart.splice(i,1);
  localStorage.setItem("cart",JSON.stringify(cart));
  loadCart();
}

function removeItem(i){
  cart.splice(i,1);
  localStorage.setItem("cart",JSON.stringify(cart));
  loadCart();
}
```

Css/style.css

```
body { font-family: Arial; margin:0; }
header { background:#333; color:#fff; padding:15px; text-align:center; }

nav {
  display:flex;
  justify-content:center;
  gap:20px;
  background:#eee;
  padding:10px;
}
nav a {
  font-weight:bold;
  cursor:pointer;
  text-decoration:none;
  color:black;
}

nav a.disabled {
  color:gray;
  pointer-events:none;
}

.form-box {
  max-width:350px;
  margin:30px auto;
  border:1px solid #ccc;
  padding:20px;
}

input, button {
  width:100%;
  padding:8px;
  margin:6px 0;
}
```

```
.link {
  color:blue;
  text-decoration:underline;
  cursor:pointer;
}

.grid {
  display:grid;
  grid-template-columns:repeat(auto-fit,minmax(200px,1fr));
  gap:20px;
  padding:20px;
}

.item {
  border:1px solid #ccc;
  padding:10px;
  text-align:center;
}

.item img {
  width:100%;
  height:150px;
  object-fit:cover;
}

.cart-box {
  max-width:600px;
  margin:20px auto;
}

.cart-item {
  display:grid;
  grid-template-columns: 2fr 1fr 2fr 1fr;
  align-items:center;
  border-bottom:1px solid #ccc;
  padding:10px 0;
```

```
}

.qty-btn {
  width:auto;
  padding:4px 8px;
}

/* --------- MEDIA QUERY FOR NAV --------- */
@media(max-width:768px){
  nav {
    flex-direction:column;
  }
}
```

Experiment 2:

AIM: Build a **responsive web application** for shopping cart with registration, login, catalog and cart pages **using Bootstrap framework**

SOLUTION:

Bootstrap is a front-end framework that provides ready-made CSS styles and JavaScript components, which we use inside HTML pages.

Make websites responsive:

Bootstrap automatically adjusts layout for:

mobile

tablet

laptop

desktop

Source Code:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>Shopping Cart</title>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.
css" rel="stylesheet">
<link rel="stylesheet" href="css/style.css">
</head>
<body>

<!-- NAVBAR -->
<nav class="navbar navbar-dark bg-primary navbar-expand-md px-3">
  <div class="container ">
  <a class="navbar-brand" href="#" id="brand">Shopping Cart</a>

  <div class="d-flex flex-column flex-md-row gap-2">
    <button      id="navLogin"      class="btn      btn-light      btn-sm"
onclick="showPage('login')">Login</button>
    <button     id="navRegister"     class="btn     btn-light     btn-sm"
onclick="showPage('register')">Register</button>
    <button             class="btn             btn-light             btn-sm"
onclick="showPage('catalog')">Catalog</button>
    <button class="btn btn-light btn-sm" onclick="showPage('cart')">
      Cart (<span id="cartCount">0</span>)
    </button>
    <button   id="navLogout"   class="btn   btn-danger   btn-sm   d-none"
onclick="logout()">Logout</button>
```

```
  </div>
</div>
</nav>

<div class="container mt-4">

  <!-- LOGIN PAGE -->
  <div id="login" class="page">
    <div class="card p-4 mx-auto" style="max-width:400px">
      <h4 class="mb-3">Login</h4>
      <input        id="loginUser"        class="form-control        mb-2"
placeholder="Username">
      <input  id="loginPass"  type="password"  class="form-control  mb-2"
placeholder="Password">
      <button           class="btn           btn-primary           w-100"
onclick="login()">Login</button>
      <p class="mt-2" id="loginMsg"></p>
      <p class="mt-2 text-center">
        New  user?  <a  href="#"  onclick="showPage('register')">Create  an
account</a>
      </p>
    </div>
  </div>

  <!-- REGISTER PAGE -->
  <div id="register" class="page">
    <div class="card p-4 mx-auto" style="max-width:400px">
      <h4 class="mb-3">Register</h4>
      <input id="regUser" class="form-control mb-2" placeholder="Username">
      <input   id="regPass"   type="password"   class="form-control   mb-2"
placeholder="Password">
      <button           class="btn           btn-primary           w-100"
onclick="register()">Register</button>
      <p class="mt-2" id="regMsg"></p>
      <p class="mt-2 text-center">
        Already       have       an       account?       <a       href="#"
onclick="showPage('login')">Login</a>
      </p>
    </div>
  </div>

  <!-- CATALOG -->
  <div id="catalog" class="page">
    <div class="row" id="productList"></div>
  </div>

  <!-- CART -->
  <div id="cart" class="page">
    <h4>Your Cart</h4>
    <div id="cartItems"></div>
    <h5 class="mt-3">Total: ₹ <span id="total">0</span></h5>
  </div>

</div>
```

```
<script src="js/app.js"></script>
</body>
</html>
```

Js/app.cs

```
// ---------- STATE ----------
let loggedIn = false;
let cart = [];

// ---------- PRODUCTS ----------
let products=[
  {id:1,             name:'Men             Shorts',             price:1199,
img:'https://images.pexels.com/photos/769749/pexels-photo-769749.jpeg'},
  {id:2,            name:'Women            Shirts',            price:1199,
img:'https://images.pexels.com/photos/3775087/pexels-photo-3775087.jpeg'},
  {id:3,            name:'Running            Shoes',            price:1200,
img:'https://images.pexels.com/photos/2529148/pexels-photo-2529148.jpeg'},
  {id:4,            name:'Sports            Shoes',            price:1500,
img:'https://images.pexels.com/photos/19090/pexels-photo.jpg'},
  {id:5,             name:'Casual             Shoes',             price:900,
img:'https://images.pexels.com/photos/2529147/pexels-photo-2529147.jpeg'},
  {id:6,              name:'Mobile              A',              price:12000,
img:'https://images.pexels.com/photos/607812/pexels-photo-607812.jpeg'},
  {id:7,              name:'Mobile              B',              price:15000,
img:'https://images.pexels.com/photos/1334597/pexels-photo-1334597.jpeg'},
  {id:8,              name:'Mobile              C',              price:18000,
img:'https://images.pexels.com/photos/699122/pexels-photo-699122.jpeg'},
  {id:9,              name:'Laptop              A',              price:42000,
img:'https://images.pexels.com/photos/18105/pexels-photo.jpg'},
  {id:10,             name:'Laptop              B',              price:55000,
img:'https://images.pexels.com/photos/374074/pexels-photo-374074.jpeg'}
];

// ---------- PAGE CONTROL ----------
function showPage(page){
  document.querySelectorAll('.page').forEach(p=>p.style.display='none');

  if((page=='catalog'||page=='cart') && !loggedIn){
    alert('Please login first');
    page='login';
  }
  document.getElementById(page).style.display='block';
}

// ---------- REGISTER ----------
function register(){
  regMsg.innerText = '';
  const username = regUser.value.trim();
  const password = regPass.value.trim();

  if(!username || !password){ regMsg.innerText='All fields are required';
return; }
```

```
  if(username.length<3){  regMsg.innerText='Username  must  be  at  least  3
characters'; return; }
  if(password.length<4){  regMsg.innerText='Password  must  be  at  least  4
characters'; return; }

  localStorage.setItem('user', username);
  localStorage.setItem('pass', password);
  regMsg.innerText='Registered successfully! Redirecting to login...';
  regMsg.style.color = 'green';

  regUser.value='';
  regPass.value='';

  // Redirect to login after 1 second
  setTimeout(() => {
    showPage('login');
    regMsg.innerText = '';
  }, 1000);


}

// ---------- LOGIN ----------
function login(){
  loginMsg.innerText='';
  const username=loginUser.value.trim();
  const password=loginPass.value.trim();
  if(!username){ loginMsg.innerText='Username is required'; return; }
  if(!password){ loginMsg.innerText='Password is required'; return; }

  const storedUser=localStorage.getItem('user');
  const storedPass=localStorage.getItem('pass');

  if(username===storedUser && password===storedPass){
    loggedIn=true;
    navLogin.classList.add('d-none');
    navRegister.classList.add('d-none');
    navLogout.classList.remove('d-none');
    loginUser.value=''; loginPass.value=''; loginMsg.innerText='';
    showPage('catalog');
  } else{
    loginMsg.innerText='Invalid credentials';
  }
}

// ---------- LOGOUT ----------
function logout(){
  loggedIn=false; cart=[];
  updateCart();
  navLogin.classList.remove('d-none');
  navRegister.classList.remove('d-none');
  navLogout.classList.add('d-none');
  showPage('login');
}
```

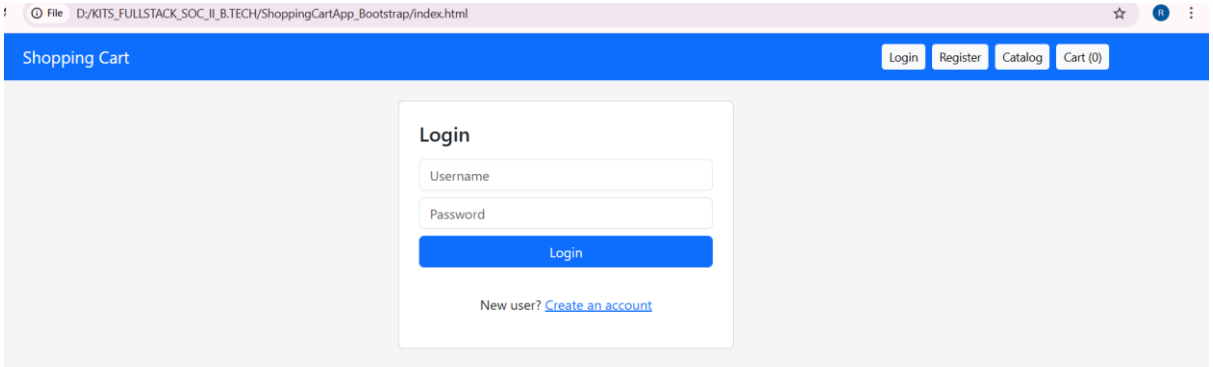
```
// ---------- LOAD PRODUCTS ----------
products.forEach(p=>{
  productList.innerHTML+=`
    <div class="col-12 col-sm-6 col-md-4 col-lg-3 mb-3">
      <div class="card text-center">
        <img src="${p.img}" class="card-img-top">
        <div class="card-body">
          <h6>${p.name}</h6>
          <p>₹${p.price}</p>
          <button          class="btn          btn-primary          btn-sm"
onclick="addToCart(${p.id})">Add to Cart</button>
        </div>
      </div>
    </div>
  `;
});

// ---------- CART ----------
function addToCart(id){
  let item=cart.find(i=>i.id===id);
  if(item) item.qty++;
  else{    let    p=products.find(p=>p.id===id);    cart.push({...p,qty:1});
alert(p.name+' added to cart'); }
  updateCart();
}

function updateCart(){
  cartItems.innerHTML=''; let total=0,count=0;
  cart.forEach((c,i)=>{
    total+=c.price*c.qty; count+=c.qty;
    cartItems.innerHTML+=`
      <div class="d-flex justify-content-between border-bottom py-2">
        <div>${c.name} - ₹${c.price} x ${c.qty}</div>
        <div>
          <button         class="btn         btn-sm         btn-secondary"
onclick="changeQty(${i},-1)">-</button>
          <button         class="btn         btn-sm         btn-secondary"
onclick="changeQty(${i},1)">+</button>
          <button          class="btn          btn-sm          btn-danger"
onclick="removeItem(${i})">Remove</button>
        </div>
      </div>`;
  });
  totalEl=document.getElementById('total');
countEl=document.getElementById('cartCount');
  totalEl.innerText=total; countEl.innerText=count;
}

function     changeQty(i,val){     cart[i].qty+=val;     if(cart[i].qty<=0)
cart.splice(i,1); updateCart(); }
function removeItem(i){ cart.splice(i,1); updateCart(); }

// ---------- START ----------
```

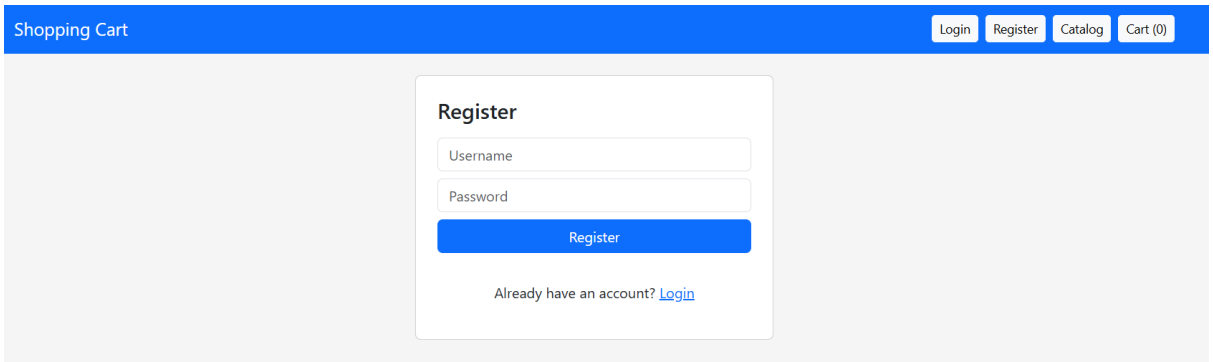
```
showPage('login');
```

<u>css/style.css</u>

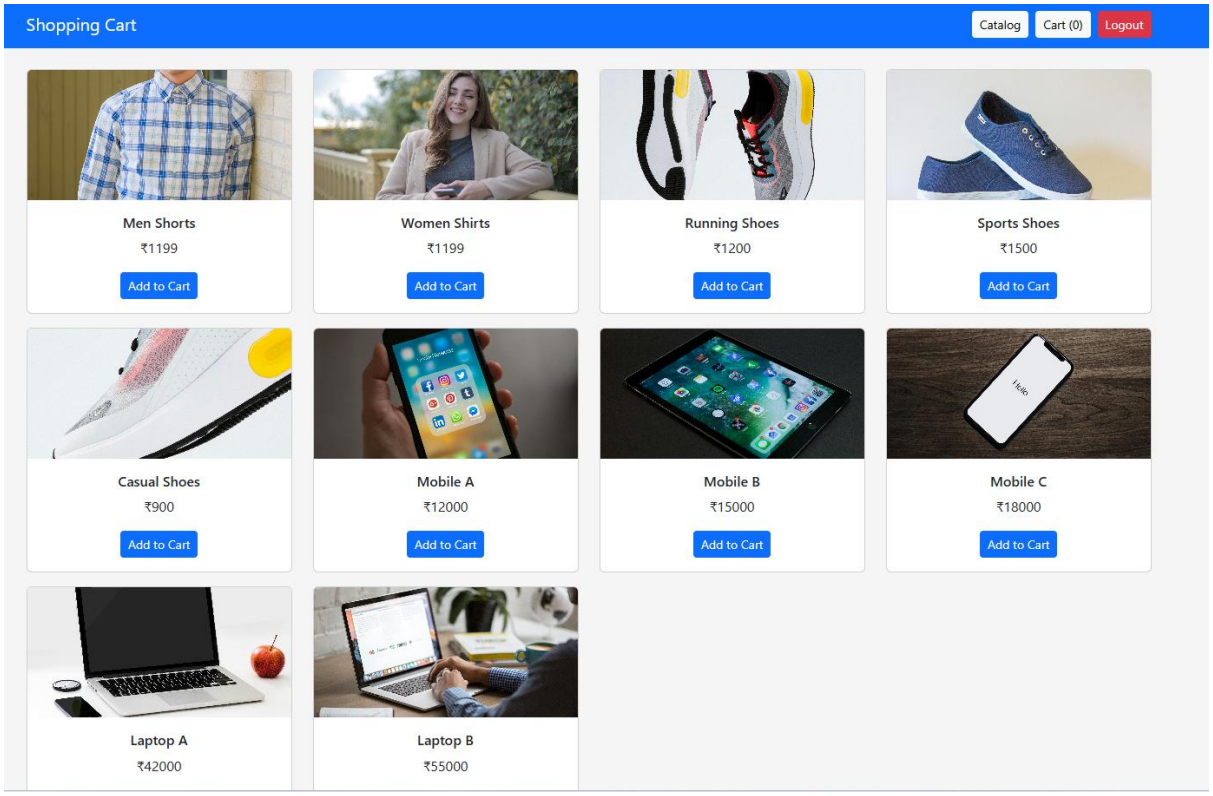
```
body { background:#f5f5f5 }
.page { display:none }
img { height:150px; object-fit:cover }
```

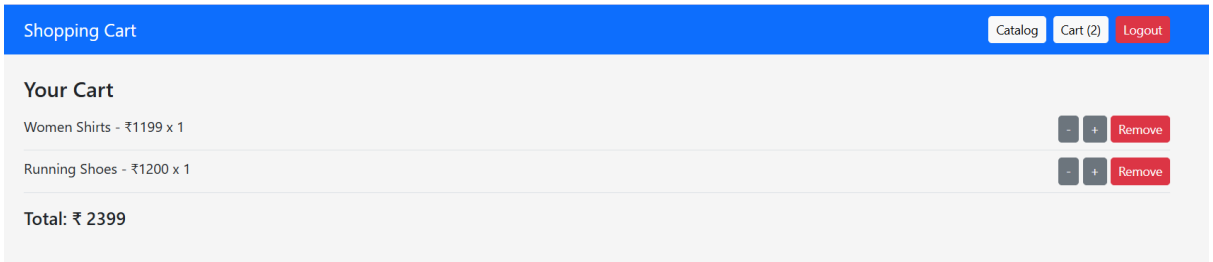
Output:



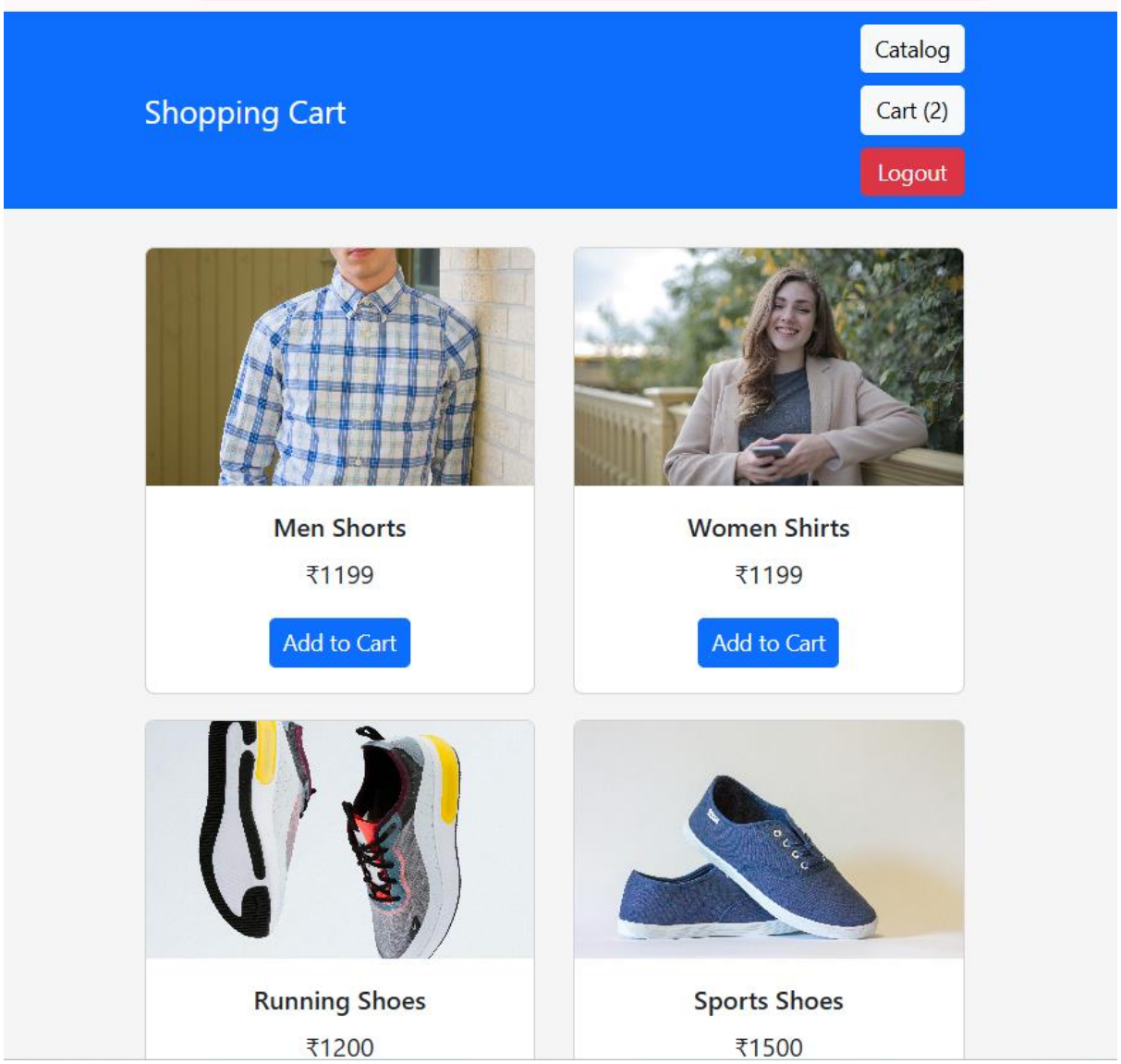


Laptop responsive web application





Tab/medium size responsive web application


Shopping Cart
Catalog
Cart (2)
Logout
Men Shorts
₹1199
Add to Cart
Women Shirts
₹1199
Add to Cart
Running Shoes
₹1200
Sports Shoes
₹1500

Mobile First responsive web application

Experiment:3

AIM: Use JavaScript for doing client – side validation of the pages implemented in experiment 1 and experiment 2.

SOLUTION:

Source Code:

Js/validation.js

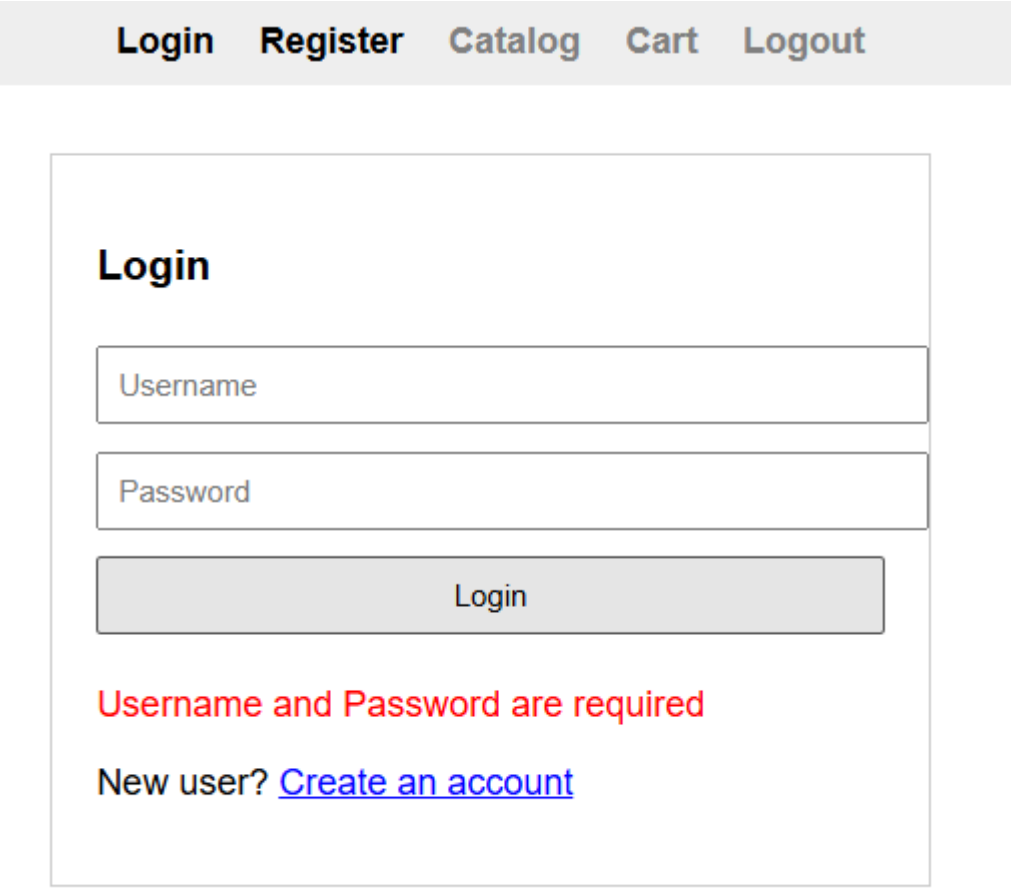
```
// ---------- REGISTER ----------
function register() {
  let username = regUser.value.trim();
  let password = regPass.value.trim();

  // ---- CLIENT SIDE VALIDATION ----
   if (username === "" && password === "") {
    regMsg.style.color = "red";
    regMsg.innerText = "Username and Password are required";

   return;
  }
  if (username === "") {
    regMsg.style.color = "red";
    regMsg.innerText = "Username is required";
    return;
  }

  if (username.length < 3) {
    regMsg.style.color = "red";
    regMsg.innerText = "Username must be at least 3 characters";
    return;
  }

  if (password === "") {
    regMsg.style.color = "red";
    regMsg.innerText = "Password is required";
    return;
  }
```

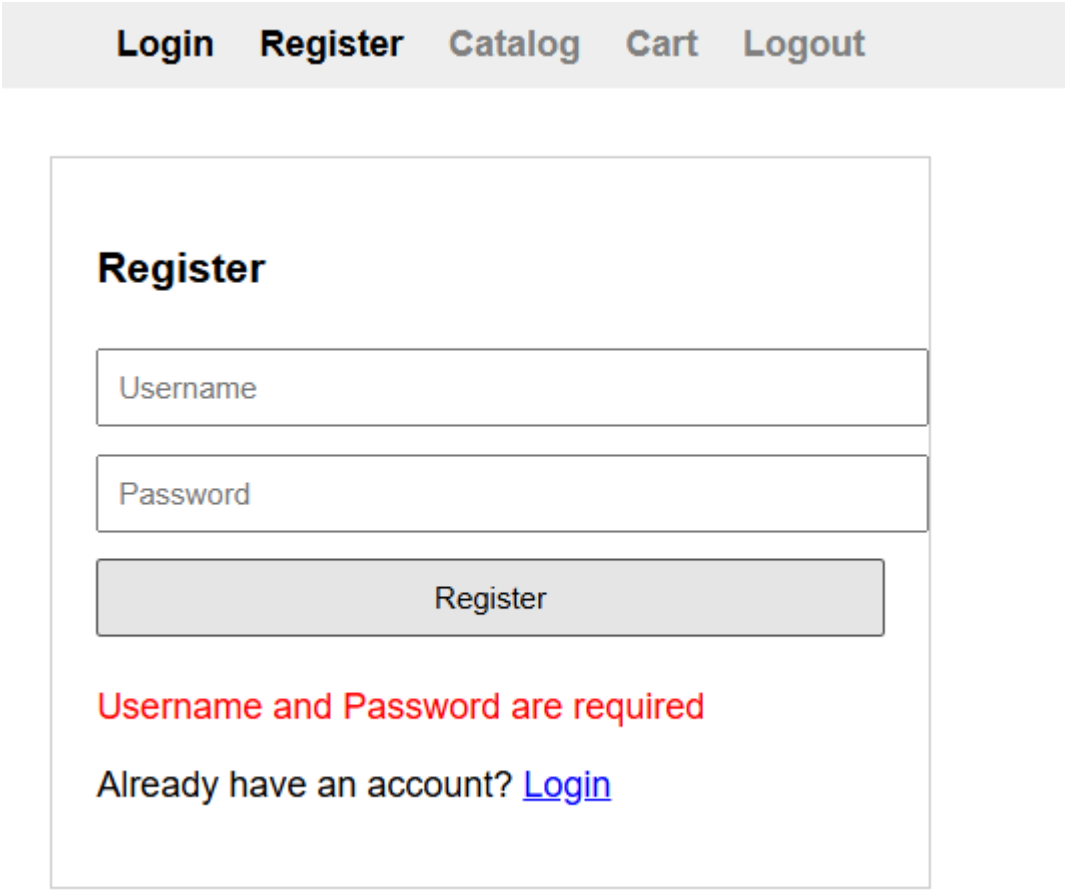
```
  if (password.length < 6) {
    regMsg.style.color = "red";
    regMsg.innerText = "Password must be at least 6 characters";
    return;
  }
  // ---- END VALIDATION ----

  localStorage.setItem("user", username);
  localStorage.setItem("pass", password);

  regUser.value = "";
  regPass.value = "";

  regMsg.style.color = "green";
  regMsg.innerText = "Registered successfully. Please login.";
  //showPage('login');
}
// ---------- LOGIN ----------
function login() {
 // ---- CLIENT SIDE VALIDATION ----
  if ( logUser.value.trim()===""  && logPass.value.trim()===""){
   logMsg.style.color = "red";
   logMsg.innerText = "Username and Password are required";
   return;
}
  if (logUser.value.trim() === "") {
    logMsg.style.color = "red";
    logMsg.innerText = "Username is required";
    return;
  }

  if (logPass.value.trim() === "") {
    logMsg.style.color = "red";
    logMsg.innerText = "Password is required";
    return;
```

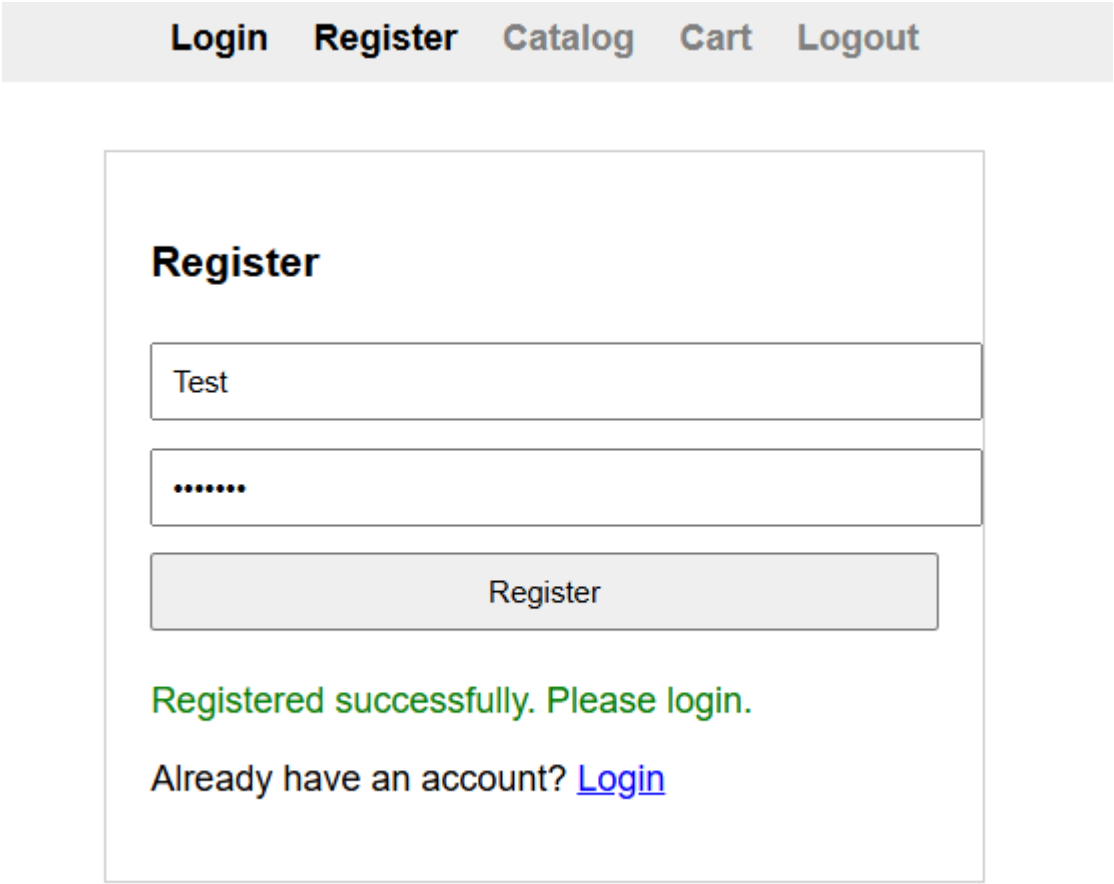
```
  }
  // ---- END VALIDATION ----

}
```

Output:

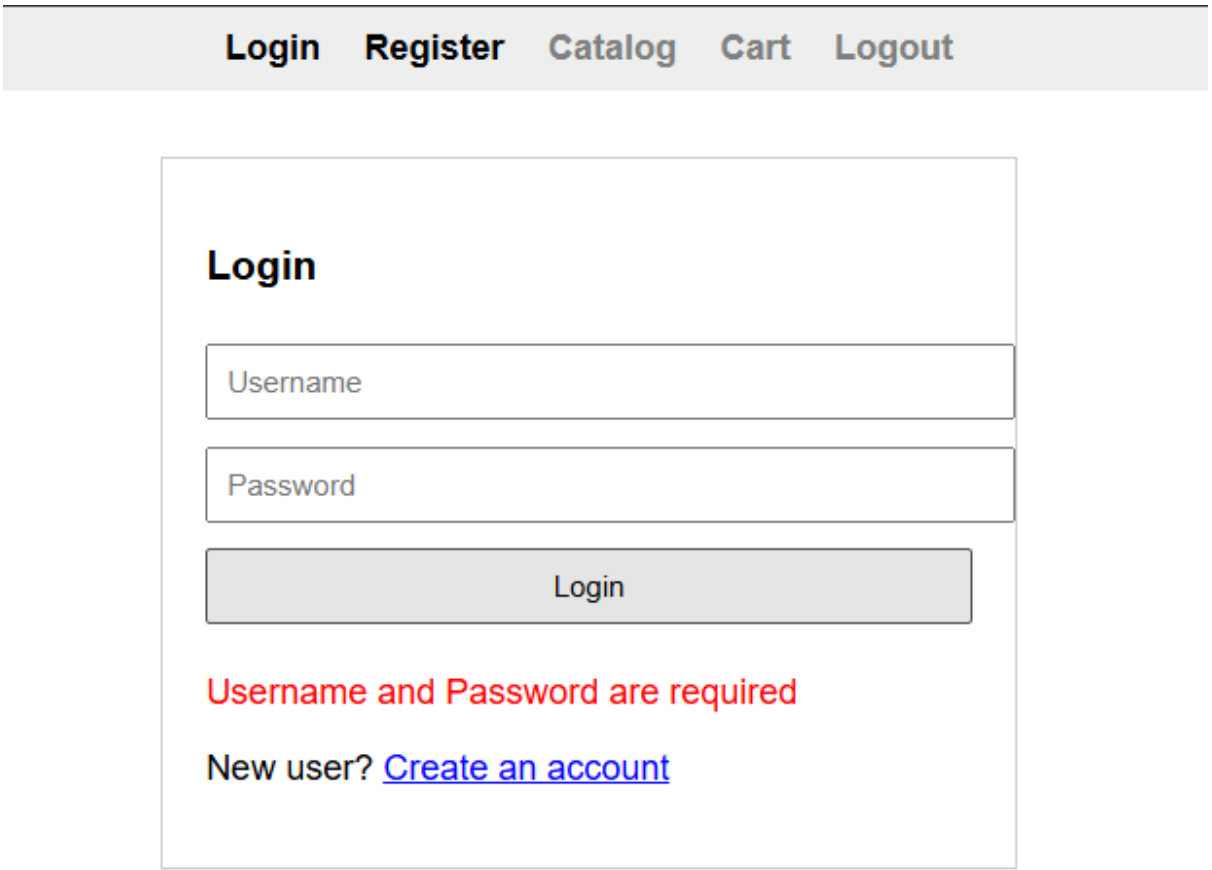
Experiment 1: Register Page Validation

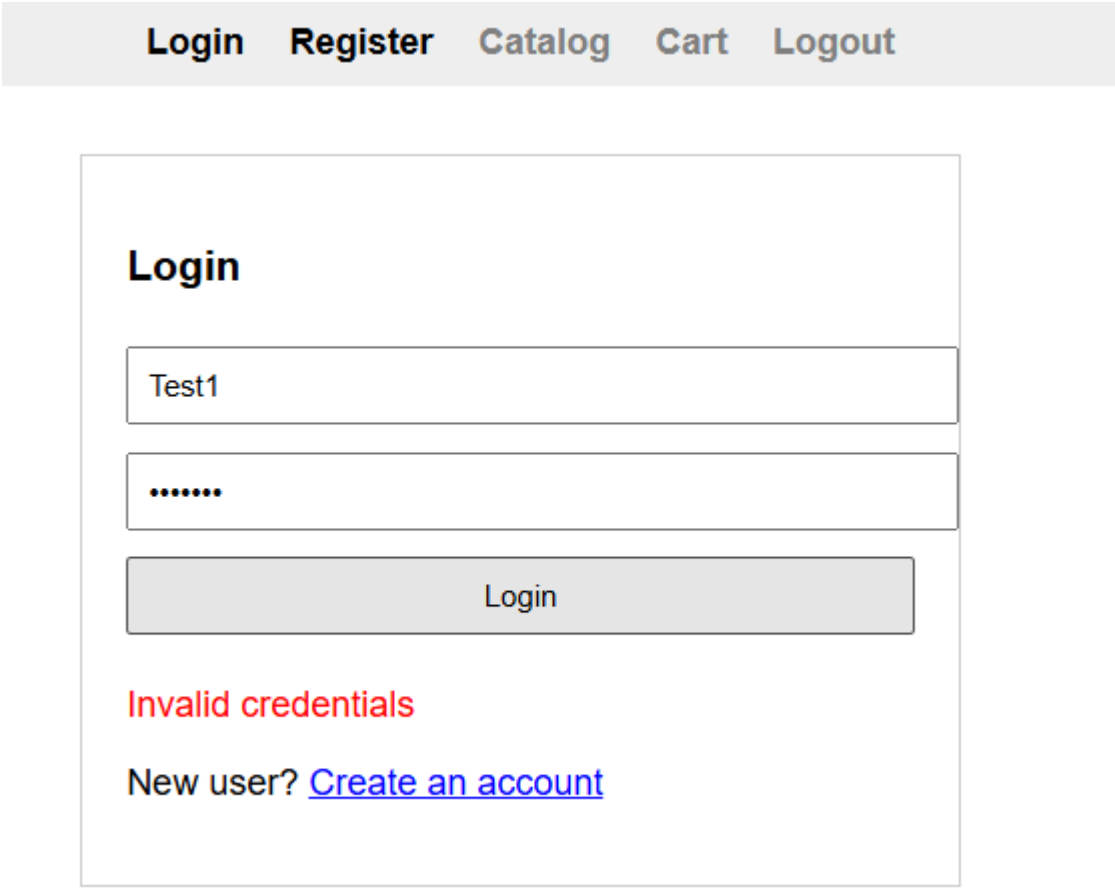


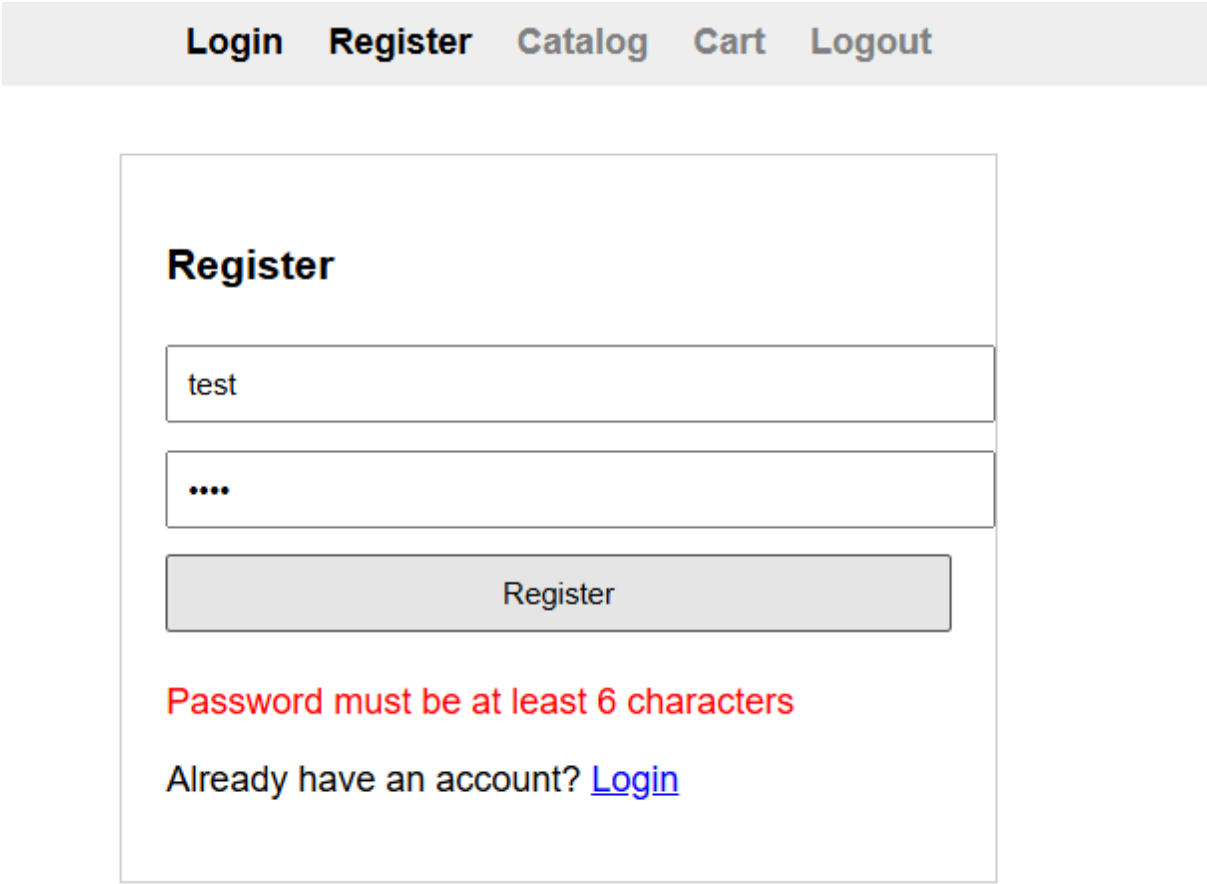

Login
Register
Catalog
Cart
Logout
Register
Username
Password
Register
Username and Password are required
Already have an account? Login


Login Register Catalog Cart Logout
Register
Test
•••••••
Register
Registered successfully. Please login.
Already have an account? Login

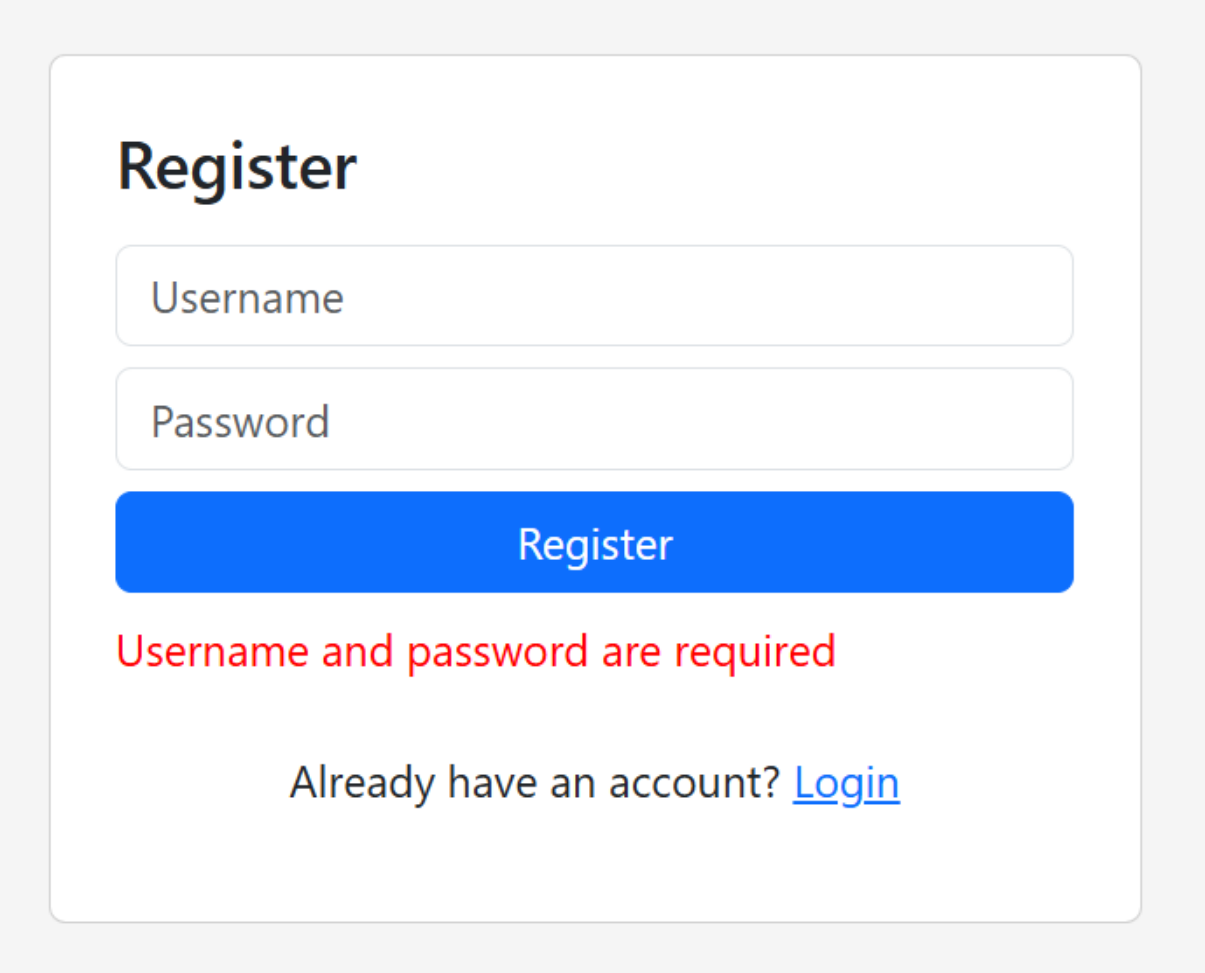
Experiment 1: Login Page Validation

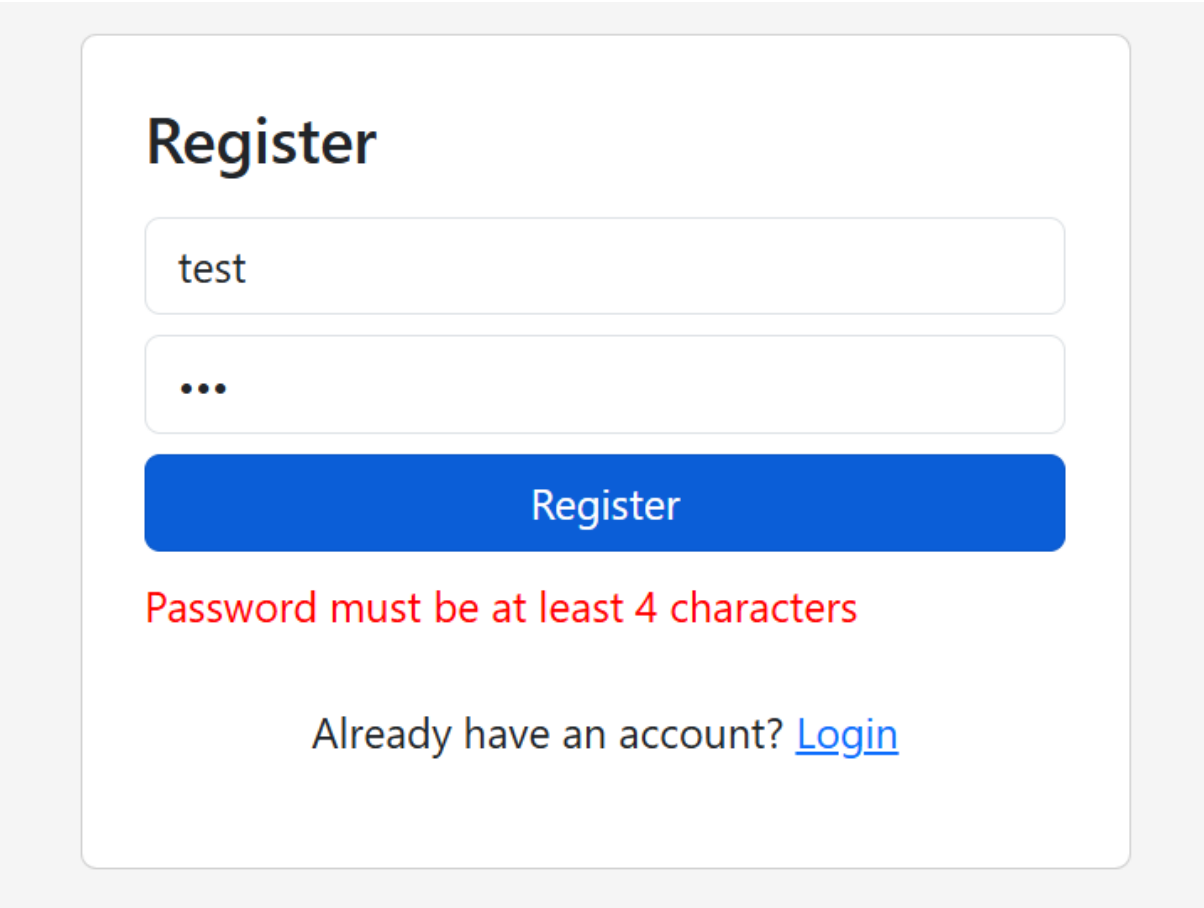


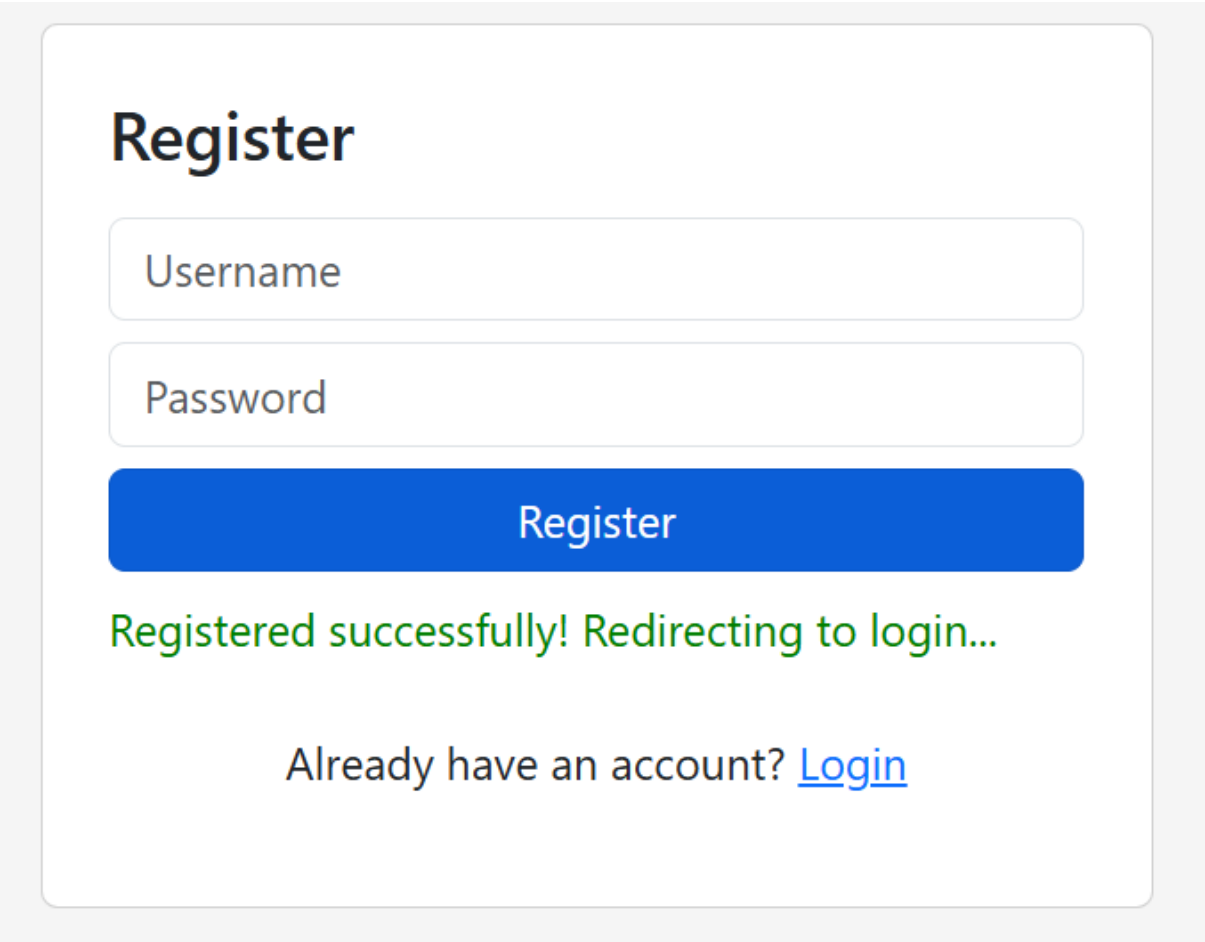

Login
Register
Catalog
Cart
Logout
Login
Test1
•••••••
Login
Invalid credentials
New user? Create an account


Login
Register
Catalog
Cart
Logout
Register
test
••••
Register
Password must be at least 6 characters
Already have an account? Login

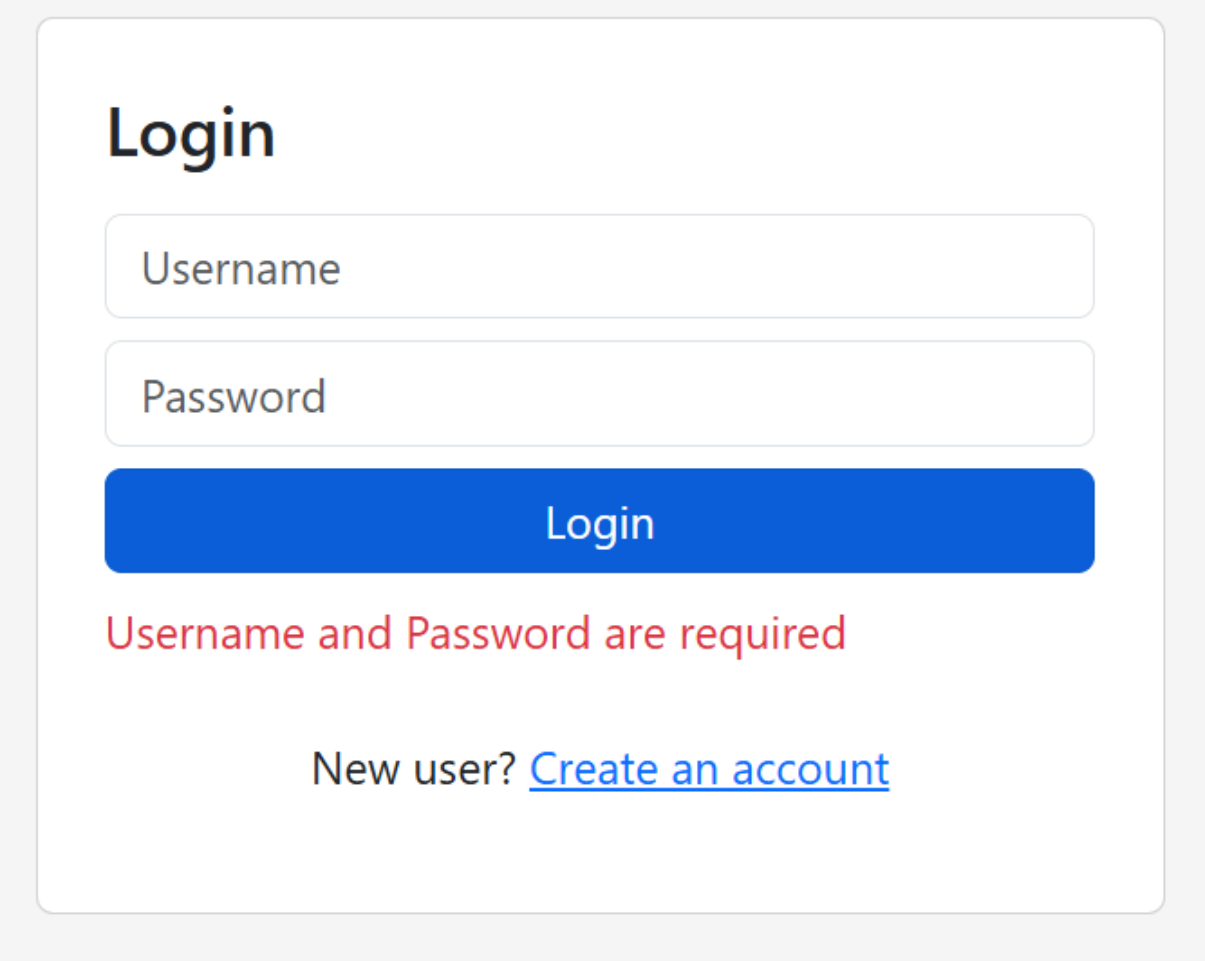
Experiment2: Registration Page Validation

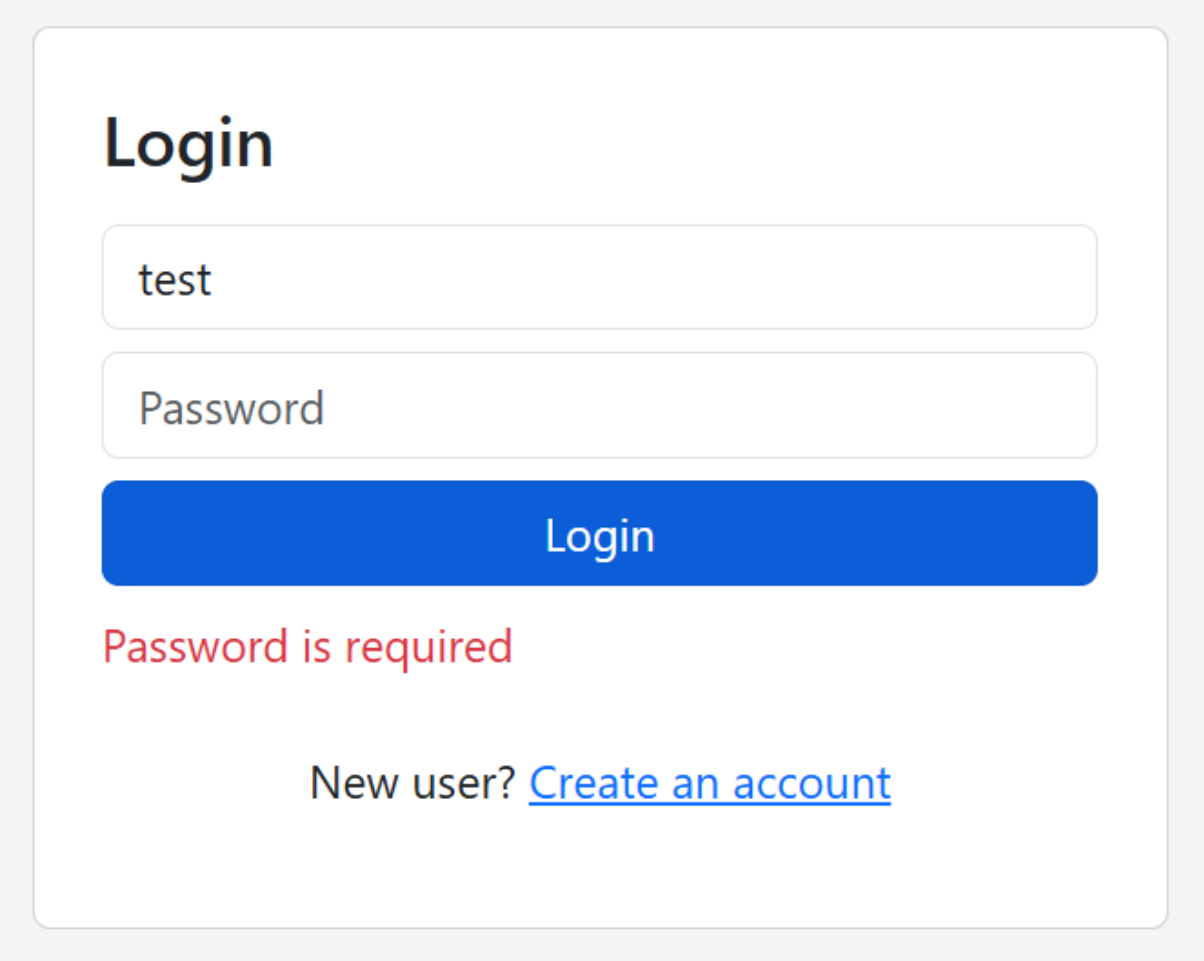


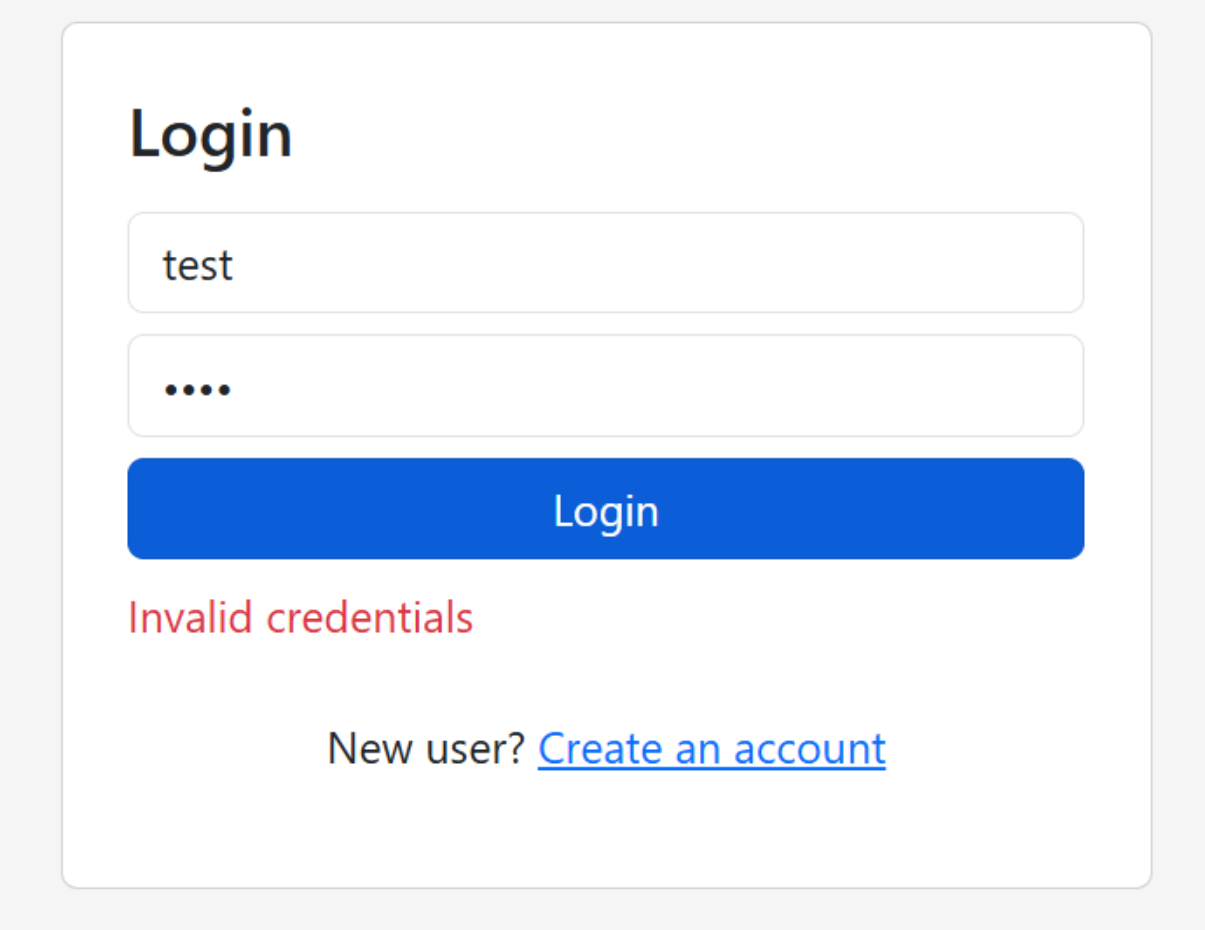

Register
test
•••
Register
Password must be at least 4 characters
Already have an account? Login



Experiment 2: Login Page Validation


Login
Username
Password
Login
Username and Password are required
New user? Create an account


Login
test
Password
Login
Password is required
New user? Create an account


Login
test
••••
Login
Invalid credentials
New user? Create an account